IT4010 Research Project – 2025 July **Topic Assessment Form**	
Project ID :	**25-26J-138**

1. Topic (12 words max)

Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

University students are increasingly facing mental health challenges, with stress and cognitive overload significantly impacting their academic performance and overall well-being [1] Traditional mental health interventions often lack scalability and accessibility, motivating the development of AI-powered chatbots to provide real-time, personalized support [2] However, existing chatbot solutions frequently rely on pattern matching or fine-tuned large language models (LLMs) that offer limited explainability and adaptability, which are critical for sensitive applications like stress regulation [3]

One major limitation of current AI chatbots is their inability to simulate human-like reasoning about emotional and psychological states. Neuro-symbolic AI, which integrates symbolic logic with neural networks, has shown promise in bringing explainability and reasoning to AI systems [4] Applying neuro-symbolic reasoning to emotion recognition can enable chatbots to interpret stress triggers more effectively and provide context-aware, empathetic responses [5]

Moreover, most chatbots focus solely on the content of user inputs and neglect behavioural signals such as typing speed, pause patterns, and lexical complexity that carry rich information about cognitive load and stress [6] Capturing such passive behavioural data can enhance the detection of cognitive overload, allowing for more timely and tailored interventions [7].

Another critical challenge is enabling chatbots to adapt continuously to individual users over time. Traditional models tend to forget previously learned patterns when updated with new data a problem known as catastrophic forgetting [8]. Continual learning techniques like Elastic Weight Consolidation (EWC) offer solutions to maintain long-term emotional state adaptation, improving user experience and personalization [9]

Finally, existing chatbots often deliver interventions at fixed intervals and in static formats, regardless of user context. This risks either overwhelming the user or providing support when it is not needed. To address this, we propose an Adaptive Multimodal Stress Intervention Timing & Modality Selector, which analyzes real-time emotional state, cognitive signals, and prior outcomes to predict both when to intervene and how (text, audio, video, etc.)—ensuring that the chatbot is not only context-aware, but also receptive-aware [10]. This adds a dynamic, human-like decision-making layer that current mental health chatbots lack [11].

This research aims to fill these gaps by developing a neuro-symbolic and cognitive AI chatbot that integrates explainable emotion reasoning, passive behavioural signal analysis, continual learning, and counterfactual stress forecasting to provide adaptive, personalized support for stress regulation among university students.

References

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6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is an AI-powered chatbot system that uses a neuro-symbolic and cognitive framework to detect, reason about, and regulate stress in university students through emotionally intelligent conversations.

The system comprises four integrated modules:

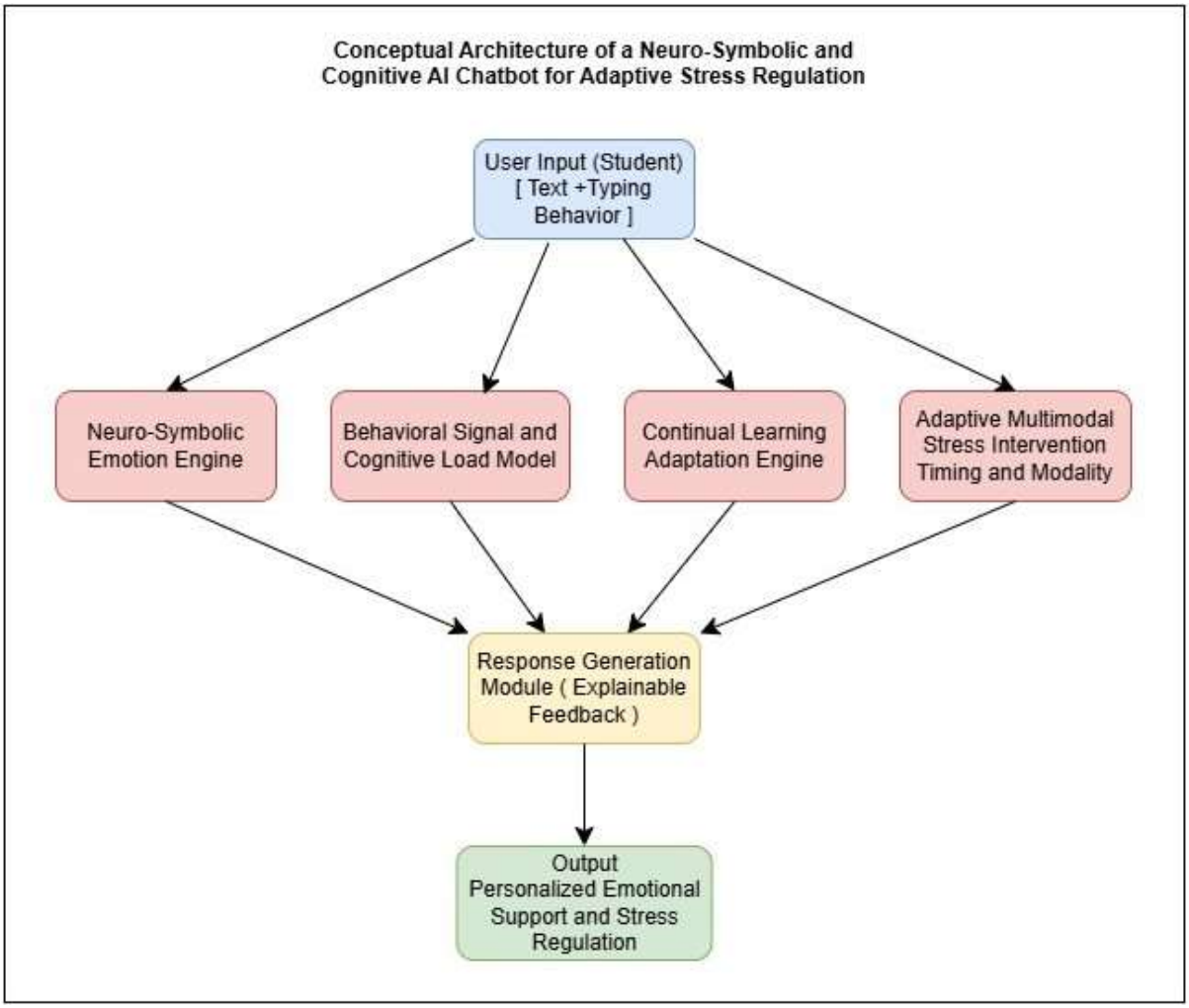
1. **Neuro-Symbolic Emotion Reasoning Engine**:
   This module combines symbolic AI (e.g., logic rules, knowledge graphs) with neural language models to understand stress-inducing triggers in conversations and It uses explainable reasoning

2. **Cognitive Load Detector**:
   Passive behavioural signals such as typing speed, hesitation pauses, and lexical difficulty are captured during user interaction. A transformer-based anomaly detection model identifies signs of cognitive overload, enabling early intervention.

3. **Continual Learning Framework**:
   The chatbot uses memory-aware learning algorithms like Elastic Weight Consolidation to learn from ongoing conversations while preserving previously acquired emotional understanding, allowing it to personalize over time without retraining from scratch.

4. **Adaptive Multimodal Stress Intervention Timing & Modality Selector**:
   Predicts the optimal moments to deliver stress-relief interventions and selects the most effective modality (text, audio, video) based on real-time emotional and behavioral signals combined with past intervention outcomes.

This holistic, explainable, and adaptive architecture offers a novel approach to delivering personalized mental health support in an ethical, scalable, and psychologically grounded manner.


Conceptual Architecture of a Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation
User Input (Student) [Text +Typing Behavior]
Neuro-Symbolic Emotion Engine
Behavioral Signal and Cognitive Load Model
Continual Learning Adaptation Engine
Adaptive Multimodal Stress Intervention Timing and Modality
Response Generation Module (Explainable Feedback)
Output Personalized Emotional Support and Stress Regulation

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research operates at the intersection of artificial intelligence, cognitive science, and psychology, requiring a multidisciplinary approach. The core expertise needed includes:

- **Neuro-Symbolic AI**: A deep understanding of hybrid models combining symbolic reasoning (ex: logic rules, knowledge graphs) with sub-symbolic learning (ex: deep learning). This is crucial for implementing explainable reasoning within the chatbot.
- **Natural Language Processing (NLP**) : Advanced NLP techniques are necessary to interpret user inputs, extract emotional cues, detect cognitive overload, and assess linguistic patterns like pause rate, lexical complexity, and sentence structure.
- **Behavioral Psychology**: insights from psychology are essential to define stress markers, emotional reasoning rules, and to interpret behavioral typing patterns for cognitive load assessment. Familiarity with psychological assessment tools and emotional models is vital.
- **Machine Learning for Time-Series Analysis**: Skills in using transformer-based models and sequence anomaly detection are needed to model user typing behavior as a temporal signal for stress or overload detection.
- **Multimodal Decision Systems**: Specialized skills in integrating multiple input streams (e.g., textual, behavioral, historical) to drive decision-making, especially in personalizing intervention timing and format.
- **Human-Centered Design**: Insight into mental health support principles and user receptiveness dynamics is needed to design appropriate and non-intrusive support strategies

Data Requirements:

- Chat transcripts labeled with emotional states and stress indicators.
- Typing behavior logs (e.g., keypress intervals, pause durations) with corresponding cognitive state annotations.
- Psychological knowledge bases (e.g., emotion-logic rules, stress-response models)

Anonymization and ethical issues are essential since mental health data is sensitive. When feasible, domain-specific data gathered through university collaborations can be combined with synthetic or publicly accessible emotional dialogue datasets (such as EmpatheticDialogues).

8. Objectives and Novelty

Main Objective

To develop an intelligent, adaptive AI chatbot for university students that goes beyond a regular chatbot and understands, monitors, and proactively regulates stress by integrating neuro-symbolic reasoning, behavioral signal analysis, continual learning, and counterfactual forecasting, enabling the system to respond empathetically, adapt over time, and simulate the impact of alternative actions for better mental well-being support.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Safwa MRF IT22628404	Design and develop a Neuro-Symbolic Emotion Reasoning Engine that can understand, interpret, and respond to university student stress in a transparent and explainable manner by combining deep learning with symbolic logic.	**1. Build the Hybrid AI Framework** • Use models like DeepProbLog or Logic Tensor Networks (LTNs) to combine deep learning with logical reasoning • Create a psychological knowledge graph that maps common student stress triggers (e.g., exams, loneliness) and coping strategies • Link emotion detection models with reasoning rules so the chatbot understands both what the student feels and why **2. Detect and Analyze Stress from Conversations** • Collect and clean sample text data from student conversations (e.g., stress-related messages) • Train models (like CNNs or RNNs) to detect emotions from text	This system uniquely combines deep learning with symbolic logic to create a chatbot that not only detects stress but also explains its actions. It's smarter than normal chatbots because: • It shows why it gives certain suggestions • It uses real psychological knowledge to guide decisions • It makes emotional AI more transparent, trustworthy, and usable in real support settings

		• Combine emotion detection with symbolic logic to understand the type and cause of stress (academic, social, etc.) **3. Generate Helpful and Explainable Responses** • Develop logic rules to suggest suitable responses *(e.g., "Exam stress + sad tone → Recommend study break and deep breathing")* • Ensure every response has a clear explanation that shows the reasoning path • Make the system learn and improve over time while keeping it easy to understand	
Rathnayake R M C T IT22332912	By passively examining users' typing habits and linguistic patterns during chatbot conversations, it is possible to identify and evaluate users' cognitive overload in real-time and enable dynamic conversational adaption to lessen mental strain.	**1. Data Collection** • Gather behavioural typing data sych as keystroke timing, pause durations, and response latency during chatbot converstaions. **2. Fearure Extraction** • Examine linguistic (lexical complexity, sentence length, grammatical structure) and temporal (typing speed, hesitations) indicators of cognitive stress. **3. Model Development** • Train cognitive load detection models using behavioral biometrics	• Introduces passive cognitive load detection using real-time typing behaviour (ex-pauses, typing speed) as a mental state indicator without requiring explicit user feedback/inputs. • Applies Sequence Anomaly Detection using transform-based models to identify deviations in user typing patterns which is a method rarely used in mental health focused chatbot systems. • Combines temporal behavioral signals with linguistic complexity

		and NLP based features. Implement transform-based sequence anomaly detection to identify stress-related deviation in typing patterns. **4. Validation** • Validatemodel accuracy using labeled datasets from controlled experiments, surveys or annotated stress benchmarks. **5. System Integration** • Integrate the cognitive load detection moduleinto chatbot's dialogue system to enable real-time adjustment of response style, pacing, and complexity based on user stress signals.	metrics to detect cognitive overload, enabling more context aware and adaptive chatbot response strategy.
Dahanaka D.M.C.B. IT22886118	Develop a continual learning system that allows the chatbot to adapt to student emotions over time by learning from new interactions without forgetting past emotional patterns.	**1. Build the Base Emotion Classifier.** • Train a BERT or LSTM model to classify emotions using datasets like EmpatheticDialogues or DailyDialog. **2. Prepare Incremental Learning Sessions.** • Divide conversation data into batches (ex: Week 1, Week 2) to simulate a student interacting over time. **3. Implement Continual Learning Methods.** • Apply Elastic Weight Consolidation (EWC) or Progress and Compress to retain previous emotional	This component introduces a continual learning system that enables the chatbot to adapt to students' emotional changes over time without forgetting past patterns. Using techniques like Elastic Weight Consolidation (EWC) and Progress and Compress, it solves the problem of catastrophic forgetting. Unlike traditional chatbots that lose earlier knowledge, this approach supports lifelong learning and emotional consistency, making it ideal for personalized mental health support.

		knowledge while learning new ones. 4. **Preprocess and Structure Data.** • Clean and organize emotional dialogue data to fit the continual learning setup. 5. **Evaluate Model Retention and Adaptation.** • Measure forgetting rate, accuracy on earlier sessions, and compare with standard fine-tuning. 6. **Integrate Emotional Memory Module.** • Store and use past emotional states to improve personalization and predictions.	
F N F Ramla IT22003928	Develop a temporal prediction model that analyzes real-time emotional and behavioral signals to determine the optimal timing and modality for delivering personalized stress-relief interventions during chatbot conversations.	1. **Data Collection and Preprocessing** • Gather datasets containing conversational exchanges annotated with emotional states, behavioral signals (typing speed, pauses), and intervention outcomes from available datasets such as the dailyDialog dataset. • Extract and preprocess features such as emotional trajectories, typing patterns, and historical modality preferences. 2. **Temporal Intervention Timing Model** • Implement sequence models (e.g., LSTM, GRU, or Transformer) to predict the optimal timing for intervention delivery.	• Predicts the best time during a conversation to give stress-relief support, instead of interrupting randomly or too often. • Chooses the best way to deliver the support (like text, audio, or video) based on how the user is feeling and behaving right now. • Combines different signals—how the user talks, types, and responded before—to make smarter decisions.

		• Train and validate the model using labeled sequences with intervention timestamps. 3. **Intervention Modality Classifier** • Build a multi-class classification model to select the best intervention format given user context. • Train on historical data linking intervention modalities with positive user outcomes. 4. **Multimodal Data Fusion** • Develop mechanisms to combine emotional, behavioural, and historical data streams for holistic decision-making. • Integrate timing and modality prediction outputs to generate unified intervention recommendations.	

9. Individual component description ofhow it is complied with the specialization.

Member Name with Registration No	Description
Safwa M R F IT22628404	**Neuro-Symbolic Emotion Reasoning Engine** This component aligns with the specialization by combining natural language processing, machine learning, and symbolic reasoning to simulate human-like emotional understanding. It involves building hybrid models using frameworks like DeepProbLog or Logic Tensor Networks to analyze student conversations and identify stress triggers. By integrating deep learning-based emotion detection with logical rules based on psychological knowledge, the system can explain its decisions in a transparent way. This demonstrates specialized knowledge in explainable AI, emotion modeling, and the application of intelligent systems for mental health support.
Rathnayake R M C T IT22332912	**Cognitive Load Detection via Passive Behavioral Signals** This component aligns with the specialization in behavioral biometrics and affective computing by analyzing users' typing behavior and linguistic patterns to infer cognitive overload. It applies advanced temporal machine learning models (e.g., transformers) and NLP techniques to passively monitor mental strain without interrupting the user, demonstrating expertise in cognitive modeling, stress detection, and human-AI interaction.
Dahanaka D.M.C.B. IT22886118	**Continual Learning for Emotional State Adaptation** This component applies advanced AI techniques to help a chatbot learn and adapt to students' emotional states over time. It aligns with the Data Science and AI specialization by using methods like Elastic Weight Consolidation (EWC) and Progress and Compress to prevent catastrophic forgetting. The project involves emotion classification using NLP, incremental training, and performance evaluation, combining both deep learning theory and practical implementation. It contributes to building intelligent, personalized systems which is a key focus in the AI field.
F N F Ramla IT22003928	**Adaptive Intervention Timing & Modality Selector** This component directly aligns with the Data Science specialization as it involves building predictive models that determine when and how to deliver stress-relief interventions based on real-time user data. It uses temporal sequence modeling (such as LSTM or Transformers) to analyze emotional and behavioral patterns over time and predict the optimal moment for intervention. Additionally, it applies classification techniques to select the most effective modality (text, audio, or video) by processing multimodal data inputs like typing speed, emotional state, and past user responses.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Samadhi	Rathnayake	
Co-Supervisor	Ms	Pubudika	Wijesundara	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.